LEF MODEL · CHAPTER 6

Chapter 6: The Fate of the Pear and the Scars of Reality — from the Pacific Basin to the Earth's Core

Based on Paper 03 — LEF Mechanism in Practice (Parts C.7–C.11, D and E)

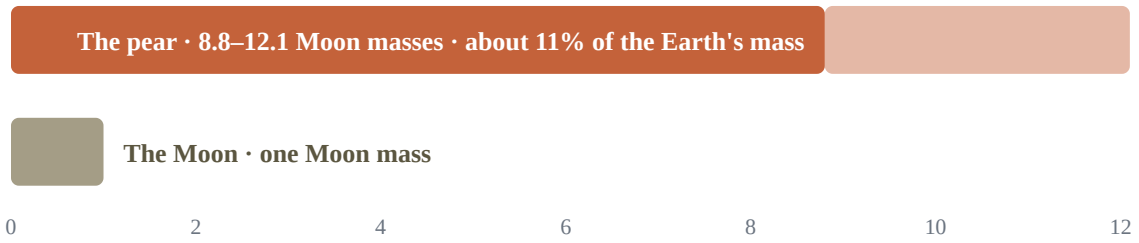
01 · THE MYSTERY

The mystery: where did 10 Moon masses disappear to — and where is the scar on the Earth's face?

In the previous chapter we saw how the Moon was released at the tip of the pear, 210 km beyond the corotation line, and soared into an independent orbit. But the moment the Moon broke away and set off, an enormous question on a planetary scale was left behind it: **what happened to the giant bridge that carried it?**

Let us remember what we are talking about. The pear that towered above the surface of the early Earth was no small hill: its base stretched across a width of **9,000 to 11,000 km**, its length from the surface to the contact disc reached about **8,700 km**, and the mass contained in it stood at **8.8 to 12.1 Moon masses** (about 11% of the mass of the whole Earth!).

Today the Earth is completely round and no giant pear towers from it into space. Where did this enormous mass disappear to? Why was it not torn off into space to make a ring around the Earth or additional moons? And most important of all — **did such a giant body, which stood on one side of the Earth for hundreds of millions of years and then sank back inward, leave behind measurable fingerprints** on the floor of the oceans, in the deep mantle and even in the Earth's core?



What was left behind: the mass of the pear compared with the mass of the Moon. All this mass lies inside the corotation line and was drawn back into the Earth.

02 · AN IMAGE

An image: a jar of sand that is shaken — and a hill of dough that is sucked back in

To understand how the pear sorted the material that built the Moon and was afterwards swallowed back into the Earth, imagine two simple experiments:

- **A jar of water and sand that is shaken and left to settle (the settling column):** take a tall glass jar filled with water, heavy grains of sand and light wood chips, and shake it well. Now stand it on the table: **gravitational sorting** begins at once — the heavy grains of sand drop down to the bottom of the jar, while the light wood chips float to the top. The whole pear lies inside the corotation line, so the gravity along it always pulls downward, toward the Earth's centre: the heavy iron sank back down into the Earth, and only a little of it reached the top and became the Moon's tiny core!
- **A wide wiper that sweeps space (why no ring formed):** imagine a giant clock hand that turns together with the Earth every 5.5–5.77 hours and reaches out to the corotation line. Any grain of dust or drop of vapour trying to orbit the Earth on a lower orbit would move faster than the pear's rotation rate — and collide with it at once! The pear acted as a cosmic wiper that gathered every particle in nearby space and prevented a ring from forming around the Earth.

Science and model: why the Moon is round and dense — and what happened to the materials that built it

Paper 03 brings together all the building components and shows how one continuous process explains the internal structure of the Moon, the small bombardment after separation, and the absence of rings:

1. **Three layers from one source (§C.10):**
   • **The crust:** the light, aluminium-rich sial layer slid along the sides of the pear on the "lubricated" layer above the sima and accumulated at the top of the dome — thereby creating an anorthositic crust rich in aluminium and thick on the far side.
   • **The mantle:** the Earth's mantle material pushed under pressure along the pear — which is why it is isotopically indistinguishable from the Earth.
   • **The tiny core:** the pear acted as a **settling column**. Most of the heavy iron that entered the pear sank back down toward the Earth, and only the little iron that managed to reach the top remained in the Moon and formed a small core with a radius of about 330 km (1%–2% of the Moon's mass, compared with about a third of the Earth's mass).

2. **Why the early Earth had no ring (§C.9a):** four reasons blocked the formation of a ring around the Earth: first, the pear itself "swept up" and gathered every particle moving on an inner orbit; second, the volatiles flowed through the pear into the Moon instead of being ejected into a ring; third, the environment was too hot for ice to condense; and fourth, right after the release, the massive Moon cleared every orbital remnant through its tidal forces and resonances. The fact that the lunar highlands preserved a saturated crater population without mechanical abrasion shows that the Moon never passed through a ring!

3. **The Earth keeps firing even after the release (§C.11):** the detachment stopped the internal pressure inside the Moon, but the openings on the Earth's surface continued to eject glowing drops and lumps of ever-decreasing size as the magma storm calmed. Measurements on the Moon's

surface confirm this in three ways: on the lunar highlands (with all the maria excluded for a fair comparison), the near side that faced the Earth carries **1.19 times** (range 1.12–1.27) more young craters 1–2 km across than the far side; the floors of the maria froze in the order of their exposure (the maria within the contact froze last and are therefore the cleanest); and the craters on the maria show a sequence of glowing bodies — large and melted at first, medium and smeared later, and small and sharp at the end.

Poincaré's paradox solved

In the classical theory of Poincaré and Darwin (1885), the whole Earth has to spin up to the brink of break-up ($\Omega^2/\pi G\rho \simeq 0.374$) in order to be stretched into a pear shape. In the LEF model, the Earth spins at a much more moderate rate ($\Omega^2/\pi G\rho \simeq 0.092$, only about a quarter of the threshold of global instability!), and the pear grows **in one local region** from a combination of rotation, density sorting and internal pressure. Since the whole pear lies inside the corotation line, no part of it stays in orbit — and all of it is drawn back into the Earth after the release!

04 · THE MECHANISM IN WORDS

The mechanism in words: the fate of the pear — how 10 Moon masses were drawn back into the Earth

From the moment the Moon broke away and went into its orbit, the enormous pear — about 8.8 to 12.1 Moon masses of Earth material (about 11% of the Earth's mass) — was left hanging entirely below the corotation line. Since at every point inside the corotation line the Earth's gravitational pull is stronger than the centrifugal push, **no part of the pear could stay in space**.

Over tens to hundreds of millions of years after the release, gravity and internal sorting gathered the whole pear back into the Earth and divided it into three clear paths (Part E of Paper 03):

1. **The heavy part sank deep inward beneath the base of the pear:** the dense part of the pear — the sima material, which carried most of the heavy iron left in it after the sorting of the settling column — sank vertically deep into the Earth's lower mantle, right beneath the region where its base had stood.

2. **The leading front was gathered into the stripes of the North Pacific:** the front part of the pear, which was stretched eastward under tension while it leaned westward, was gathered back to the surface and created the geological stripe system of the North Pacific (to be explored in the chapters on the stripes).

3. **The trailing back was gathered into the arcs of the West Pacific:** the rear part of the pear, which leaned westward toward Asia and Japan, was swallowed inward and created the island arcs and trenches of the West Pacific.

What did the pear do to the early Earth's core while it stood? (§E.5)
During the hundreds of millions of years in which the pear stood on one side of the Earth (much of the 600-million-year window up to the detachment 3.873 billion years ago), its enormous mass exerted a constant, one-sided gravitational field that reached all the way to the Earth's centre! At that early stage, the Earth's core was not yet sharply separated, but was **a wide, semi-dense "proto-core"**. The net gravitational pull of the pear at the Earth's centre — with a strength of **0.378 to 0.517 m/s²** — pulled and stretched the dense proto-core toward the pear's side! Only after the pear was swallowed back and the mantle reached isostatic balance (like a raft settling in water) did the proto-core relax and return to the centre — but it may have left an east–west signature in the Earth's crystallising inner core.

05 · THE KEY EQUATIONS

Academic explanation and the key equation

The net gravitational field acting at the Earth's centre while the pear stands (g_{center}) is calculated as the sum of the pull of the pear (g_{pear}) and the pull of the Moon (g_{Moon}), minus the centrifugal acceleration that arises because the system's common centre of mass is offset from the Earth's centre toward the pear (a_{cent}):

$$g_{\text{center}} = g_{\text{pear}} + g_{\text{Moon}} - a_{\text{cent}} = 0.378 \text{ to } 0.517 \text{ m/s}^2$$

$$W_{\text{pear}} \simeq 8.1 \times 10^{30} \text{ to } 1.05 \times 10^{31} \text{ J} \sim E_{\text{core-formation}}$$

The work of lifting the whole pear (8.1×10^{30} to 1.05×10^{31} joules, 5.4 times the work of lifting the Moon alone) matches in order of magnitude the gravitational energy released when iron sinks and builds the Earth's core (~ 10^{31} joules). When the pear sinks back inward, this potential energy is released back as heat into the Earth's thermal history.

06 · THE SIGNATURE

The signature in reality: the imprint beneath the Pacific Ocean (the LLVP province)

Modern seismic tomography — which maps the Earth's interior with earthquake waves, much like a medical CT scan — has discovered two enormous "continents" at the bottom of the Earth's mantle, right above the boundary between the mantle and the core (at a depth of about 2,800 km): broad regions in which seismic shear waves travel more slowly than average, called **large low-velocity provinces (LLVPs)** — one beneath Africa, and one beneath the Pacific Ocean.

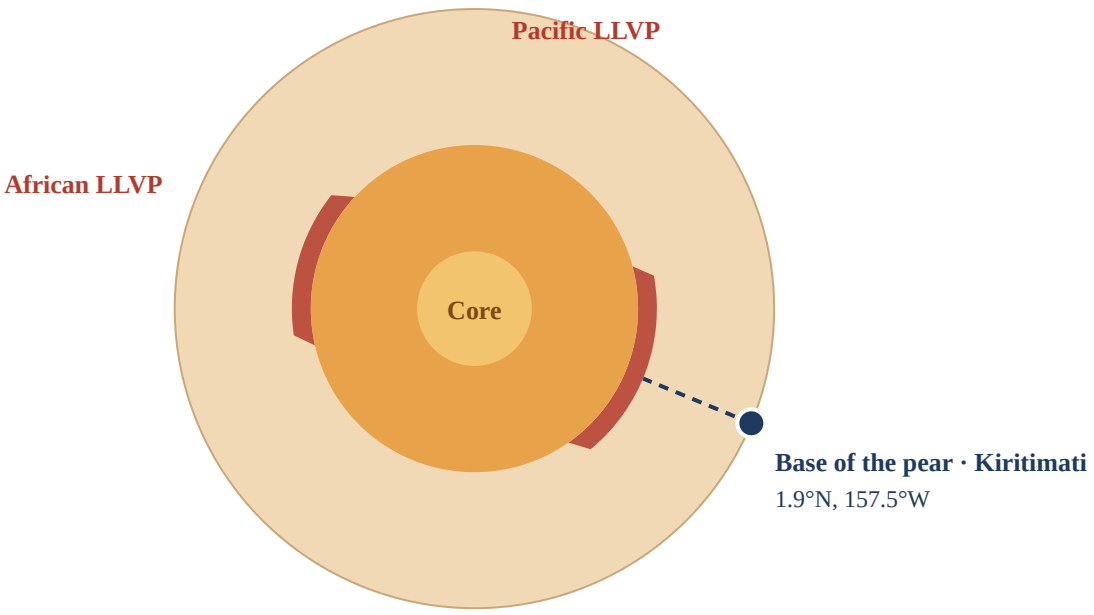
When we check where the reconstructed base point of the pear lies — near the island of **Kiritimati** in Kiribati (1.9°N, 157.5°W or 202.5°E) — on the global tomographic maps of the lower mantle (§E.4), a clear-cut result emerges:

- **Full agreement of 5 out of 5 models:** in the cluster analysis of Cottaar and Lekić (2016) at a depth of 2,550–2,850 km, all five seismic models unanimously indicate that the base point lies deep inside the slow Pacific province.
- **Deep inside the contour (−1%):** in the global mean models SMEAN and SMEAN2 at a depth of 2,800 km, the slowing of shear waves at this point is **−2.25%** and **−2.36%** respectively — and it stands more than 1,229 to 1,584 km from the edge of the province (far from any chance boundary), about 1,500 km north-north-east of the centre of the province, and about 1,120 km

(at the core–mantle boundary) from the iron-rich ultra-low-velocity zone (ULVZ) beneath Hawaii.

- **The geochemical prediction for plume basalts:** while the impact theory proposes that the mantle remnants of the foreign "Theia" sank to the bottom of the Earth and created the LLVP, in the LEF model the whole pear is made **of the Earth's own material** — so the lava rising from the mantle plumes beneath the Pacific Ocean is expected to carry a purely terrestrial isotopic fingerprint, with no foreign component from another planet!

The lab box: the balance of the pear, the settling column and the position of the base in the mantle (Paper 03, Parts C–E)



Schematic equatorial section of the Earth, seen from the North Pole

The two slow provinces (LLVPs) at the bottom of the mantle, and the reconstructed base of the pear above the Pacific one — inside it in all five models. (Schematic drawing, not to exact geographic scale.)

Component / measure	Computed or measured value	Physical and observational meaning
Dimensions and mass of the pear	Base 9,000–11,000 km, length ~ 8,700 km, mass **8.8–12.1 Moon masses**	About 11% of the Earth's mass; entirely inside the corotation line and therefore drawn back inward

Work of lifting the pear	8.1×10^{30} to 1.05×10^{31} J	5.4 times the lifting of the Moon; matches the energy of iron sinking into the core (~ 10^{31} J)
The parent's rotation parameter	$\Omega^2/(\pi G\rho) \simeq 0.092$	About a quarter of Jacobi's instability threshold (0.374) — a local, pressure-driven pear, not a global tearing
The young craters after detachment	**1.19 times** (1.12–1.27) more on the near side (1–2 km)	The last hot ejections from the Earth struck mainly the side of the Moon that faced it
Position of the base (1.9°N, 157.5°W)	**5 out of 5 models slow ($\delta V_S = -2.25\%$ to −2.36%)**	**Deep inside the Pacific LLVP province (>1,229 km from the edge at a depth of 2,800 km)**
The pull of the pear on the proto-core	$g_{\text{center}} = 0.378$ to 0.517 m/s^2	Stretched the proto-core toward the pear during building; relaxed when the pear returned

07 · NEXT

The door ahead

We have completed the mechanical decoding of the Moon's birth: we saw how a pressure of 150.26 GPa lifted foamed magma from within (Chapter 3), how the sial crust rose and divided into three bands from outside (Chapter 4), how the 3,000-km contact disc opened in the Stefan release 210 km beyond the corotation line (Chapter 5), and how the whole pear was drawn back into the Earth and left its imprint beneath the Pacific Ocean (Chapter 6).

But the moment the Moon broke away and went free is only the beginning of its long journey. **How did a moon released at latitude 22.5°, only 17,496 km from the Earth and with an initial orbital inclination of 22.5°, reach a distance of 384,400 km with an orbital tilt of only 5.145° to the plane of the ecliptic?** In the next chapter — **Chapter 7: The Journey to the Orbit — the Reduced Tangent Law and the Solution of the 5.145° Tilt** (from Paper 04 — *The Orbit*) — we will follow the Moon out into space!

Chapter 7

The Journey to the Orbit — the Reduced Tangent Law and the Solution of the 5.145° Tilt →

← Chapter 5 · Prologue

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